VISUAL RESUME

Author: Juan Mª Carvajal Ramos
Naval Architect and Marine Renewable Energies Engineer

RESUME

Hi, my name is

Juan Maria Carvajal Ramos

and I am a Naval Architect and
Marine Renewable Energies Engineer

RESUME

NAVAL ARCHITECT AND MARINE RENEWABLE ENERGIES ENGINEER

MARCH 2019

EDUCATION AND TRAINING

May 2005 - **Master's Degree in Naval Architecture and Ocean Engineering**
at Universidad Politécnica de Madrid (ETSIN-UPM)
Study areas: Ship theory, shipbuilding and marine structures

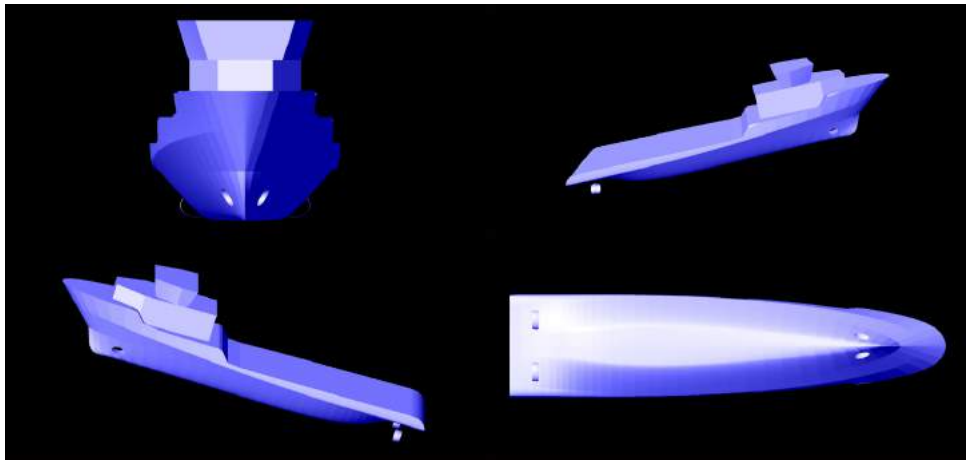
Master's thesis: ROV Offshore Support Vessel

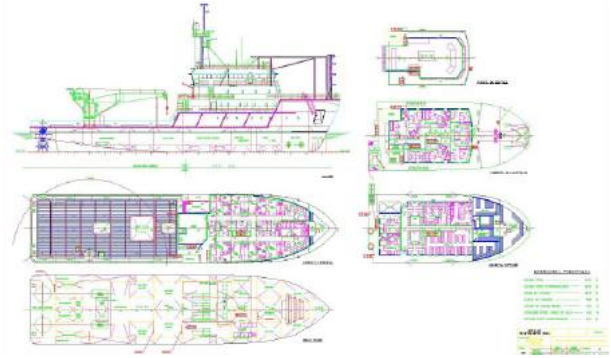
VESSEL PARTICULARS:

ABS, +A1, Circle E, +AMS, +DPS2
IMO Guidelines for vessels with DP Systems
IMO Code of safety for special purpose ships

EDUCATION AND TRAINING

LENGTH, OVERALL: 60 m
LENGTH, B.P: 55 m
BREADTH, MLD: 13.3 m
DEPTH: 5.0 m
DRAFT: 4.04 m
WETTED AREA: 740.4 m2
DISPLACEMENT: 1444 t
PRISMATIC COEFFICIENT (Cp): 0.664
BLOCK COEFFICIENT (Cb): 0.438
POWER (BHP): 2080 kW
SPEED: 10 kn

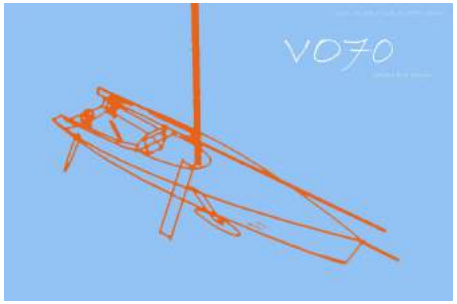




EDUCATION AND TRAINING

January 2005 - **Post-Degree of Sailing Boat Design (online)**
July 2006 at Universidad Politécnica de Madrid (ETSIN-UPM)
Study areas: Concept design, structural design, 3D modelling and hydro- and aerodynamics performance

Post-Degree's Thesis: Volvo Ocean 70 sailing boat design

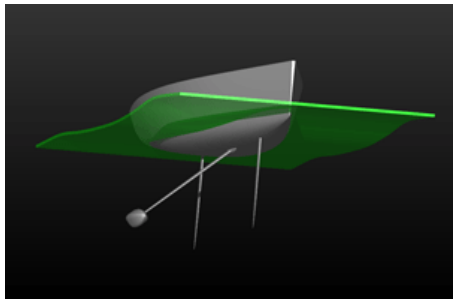


METHODOLOGY

Design development (high stability and VPP & CFD performance) including a canting keel system and a choice of daggerboards

Volvo Open 70 Rule 2003

".. .fast, single mast monohull keel-boat of similar performance, suitable for long distance racing offshore at the highest level of the sport ..."

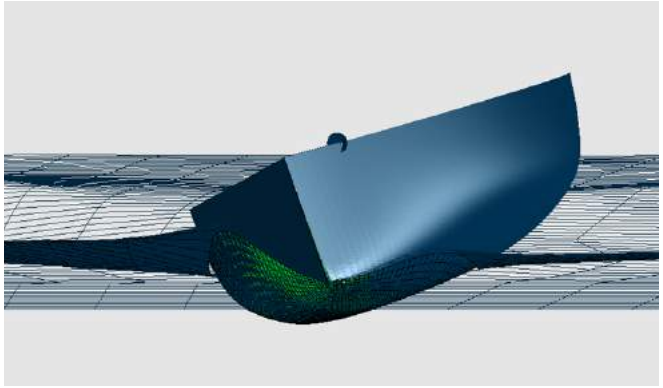


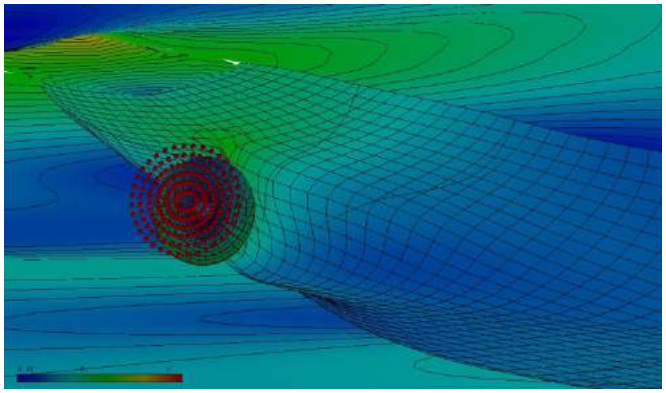
EDUCATION AND TRAINING

October 2005 - **Study abroad – Mechanical Engineering** (one year)

August 2006 at Institut für Land- und Seeverkehr im Bereich Schiffs- und Meerestechnik, Technische Universität Berlin (TU Berlin)

Study areas: Ship design, 3D modelling, hydro- and aerodynamics (CFD)





EDUCATION AND TRAINING

February 2007 - **Diploma of Advanced Studies (DEA)**

November 2011 PhD programme at Department of Naval Architecture & Marine Engineering, at Universidad Politécnica de Madrid (ETSIN-UPM)
Study areas: Ship motions (seakeeping), hull lines Optimisation (CFD)

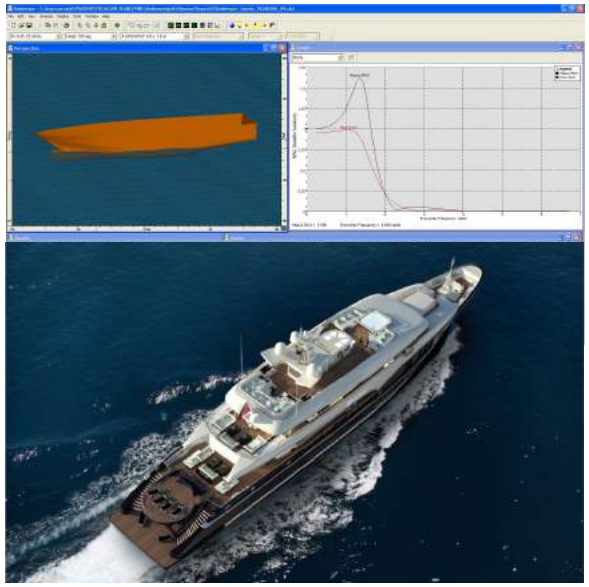
Research project: Seakeeping performance assessment of mega-yachts

METHODOLOGY

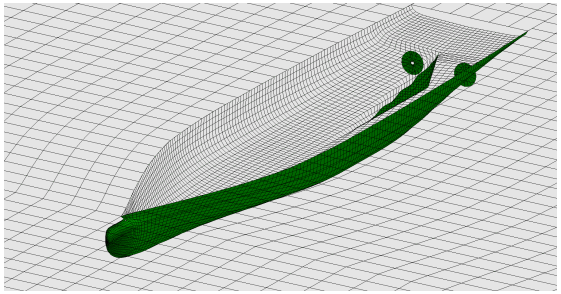
Using the strip theory to predict the vessels heave and pitch response with 3D-software Maxsurf (Seakepeer)

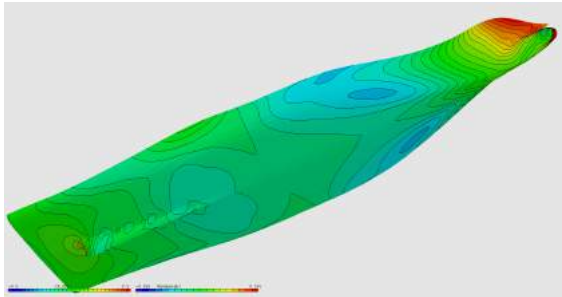
Seakeeping analysis:

- Environmental conditions (wave spectrum)
- Response characteristics of yachts (RAOs)
- Approaches used to assess the yacht's seakeeping behaviour
- Assessment of the bulbous bow with slamming criterion
- Green water loading analysis and propeller inmersion
- Calculation of vertical and lateral motions
- Operational criteria: RMS, MSI, SM, MII, LFE,...



EDUCATION AND TRAINING

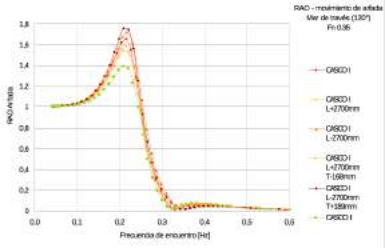


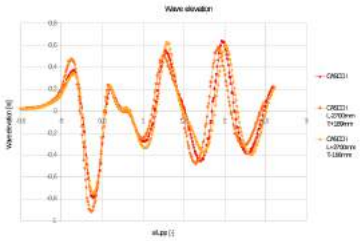


Research project: Seakeeping performance assessment of mega-yachts

Seakeeping analysis:

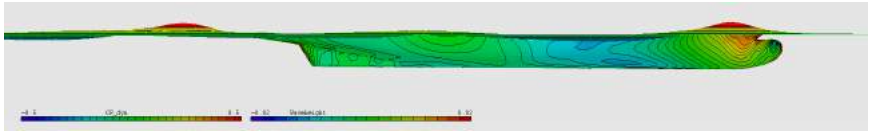
- Seakeeping assessment of series of hulls
- Slamming analysis of a specific case

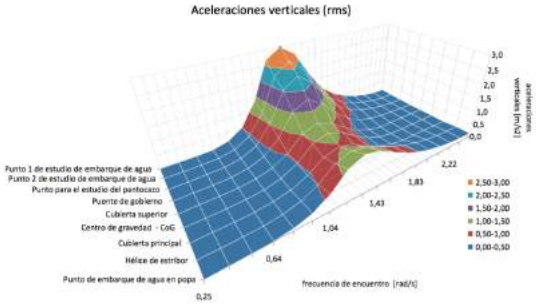




Optimisation analysis (CFD):

- Geometry definition and CFD-based hull form optimization
- FS-Flow: Non-linear free-surface Rankine panel code
- Cp, wave height, wave pattern
- Seakeeping analysis (lengthening,...)





EDUCATION AND TRAINING

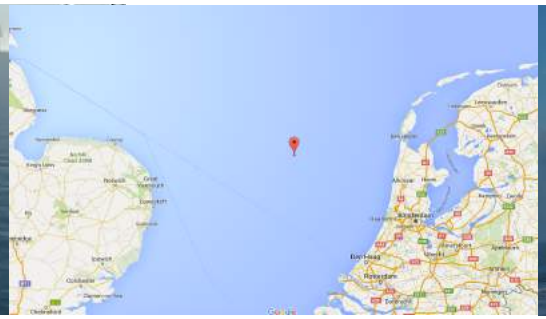
May 2015 - April 2016 **Master in Marine Renewable Energies (online / English)**

at Bureau Veritas Business School and Instituto Marítimo Español (IME)
Study areas: Technical aspects of marine technologies: Offshore Wind Energy, Projects permitting process, Shipping & Logistics, Economics, O&M, Technical aspects of other marine technologies: Wave, Tidal and Current

METHODOLOGY

- Course structured in 7 academic modules, including discussions and case studies.
- Development of a Final Project and an International Business Project

Master's Thesis: Offshore Resources Assessment – Integration of Offshore Wind and Wave Energy Resources



EDUCATION AND TRAINING

February 2019 - **Project Management Professional (PMP®) Exam**

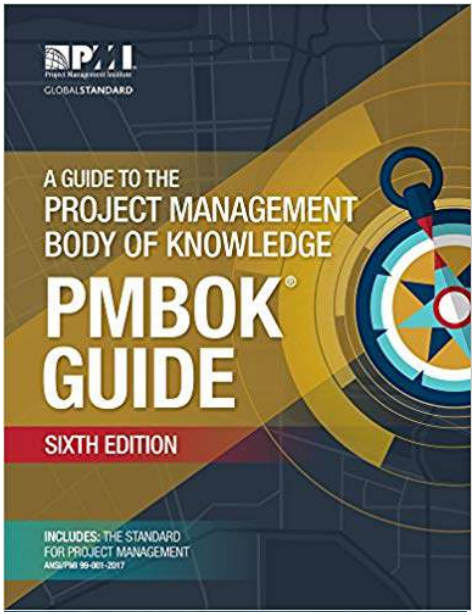
Sep. 2019 (exp.) **Preparation Course (online / Spanish & English)**

at Universidad Nacional de Educación a Distancia – U.N.E.D.
Study areas: Technical Project Management, Strategic and Business Management and Leadership

METHODOLOGY

- By taking an analytical and practical approach, this course will provide trainees with the concepts, theories, techniques and tools related to project management as described in the A Guide to the Project Management Body of Knowledge (PMBOK® Guide) – Sixth Edition, Project Management Institute (PMI), Inc.

- The course will examine all the ten knowledge areas to achieve the objective of presenting a complete and exhaustive view of the processes that must be implemented for a successful project management activity.



WORK EXPERIENCE

July 2000 -

Internship as a supernumerary on board

of the MV La Rabida, a bulk carrier, ERSHIP SAU

Summer 2002 -

Employee

Vindius Aventura SL, 33500 Llanes, Asturias

Main Activity: Group guide in climbing, canyoning, speleology & kayaking

March 2004 - June 2005

Junior Project Engineer (Internship)

at Grupo SANJOSE SA, Madrid

Main activity: Studies and Production in the hydraulic engineering depart.

January 2007 - May 2007

Junior Project Engineer

at Siport21, S.L. - Port Engineering and Maritime Safety Consulting

Main activity: Ship dynamics studies (Passing ship / Mooring projects)

WORK EXPERIENCE

May 2007 - January 2009

Junior Naval Architect and Yacht Designer

at Newcruise Yacht Projets & Design, Hamburg

Naval Architect in Exterior Design department (focused on Yacht Design)

Main Activity:

- Conceptual design and engineering on yacht project
- Development of design solutions and presentation to the client
- Ship theory and preliminary calculation of resistance and powering
- 3D modelling of hull, appendages and superstructure
- Technical drawings and layout: GAP, Hull lines plan, Safety Plan, …
- Application of Classification Rules and SOLAS, COLREG & MCA
- Seakeeping performance assessment of a mega-yacht (Research project)
- Product research and coordination with suppliers

WORK EXPERIENCE

January 2009 - October 2015

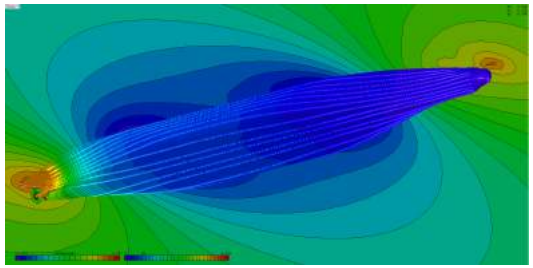
Project Engineer (R&D and CFD Engineer)

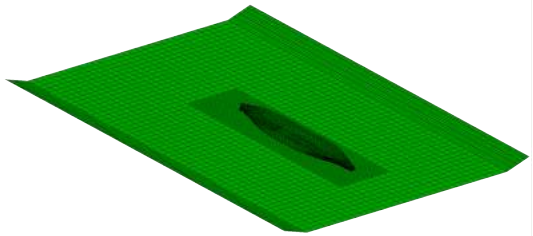
at DNV GL Maritime Advisory, Potsdam (Germany)

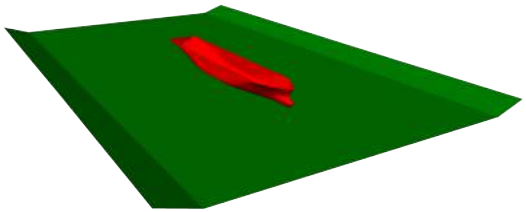
Naval Architect in Exterior Design department for projects focused on hydrodynamic optimisation using CFD simulations (Research & Development)

Main Activity:

- Hydrodynamic optimisation to enhance the vessel's operational efficiency
- R & D of specific technological projects, i.e. optimisation tool ECO Assistant
- Simulations in model / full scale using flow codes (Fine/Marine, FS-Flow)
- Parametric modelling of hull & appendages, and design solutions



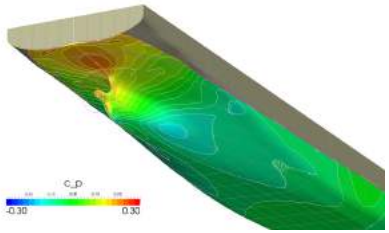




Hydrodynamic optimisation – DNV GL Maritime Advisory

Hull Optimisation (around 7 years experience):

- Hydrodynamic performance of a vessel with the study of the ship resistance and propulsive efficiency
- CFD Simulations in full scale using FINE/Marine (recognized 6DOF incompressible flow solver) and FS-Flow (Rankine source panel code)

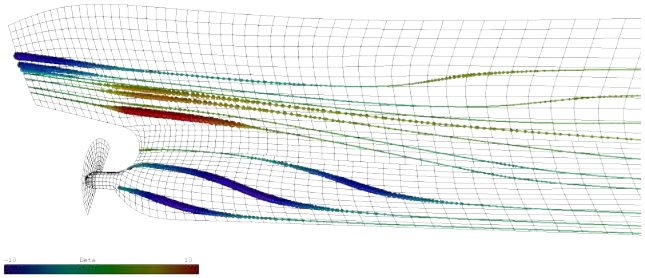


Scope of work included:

- Optimising the hull forebody and aftbody based on the operating profile
- Analysis of the wake and propulsion efficiency
- Predicting the speed – power and fuel oil consumption curves
- Assessing the preliminary EEDI
- Attending the towing tank model test

Research projects:

- Meshing studies
- Using a potential flow code to evaluate the squat effect in shallow water (Boundary layer)
- CFD evaluation of PanMax Widening Concept
- Design assessment of appendages:
    - CFD simulations without and with anchor
    - Effect of a rudder on a bare hull



Effective trim optimisation – DNV GL Maritime Advisory

ECO Assistant (around 4 years experience):

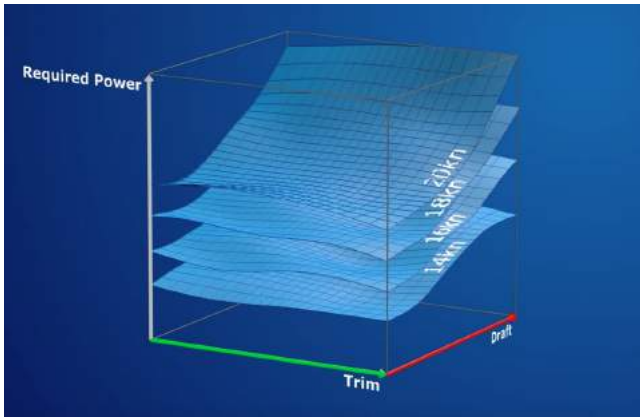
– The pressure to lower fuel consumption is stronger today than ever before.

– The hydrodynamic efficiency of a vessel may be affected by the trim. Awareness of the vessel performance in different trimmed conditions is therefore an essential step towards improving the operational efficiency and reducing the fuel consumption and corresponding emissions to air.

– One choice to assess of the effect of trim on hull resistance and fuel consumption.

Scope of work included:

– ECO Assistant is an easy-to-use, stand-alone software tool. Providing optimum trim under all operating profile. It saves tons of fuel and CO2, by delivering the optimal trim angle for a specific ship based on a few operational parameters.



– ECO Assistant calculates the optimum trim for the specifying operating condition once the user enters the operational parameters such as speed, displacement and water depth, and displays it together with the corresponding static trim.

Effective trim optimisation – DNV GL Maritime Advisory

- The ship-specific resistance data for the vessel is generated by state-of-art CFD simulation tools.

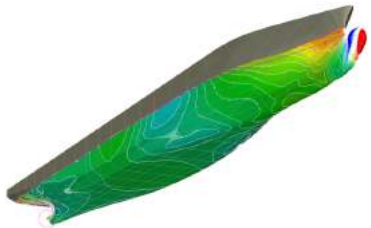
- Free-surface RANSE flow code assesses the dynamic floating condition, wave resistance from pressure integration and the viscous resistance.
- Simulations in full scale using FINE/Marine (a viscous flow code dealing with RANSE in the entire flow domain)

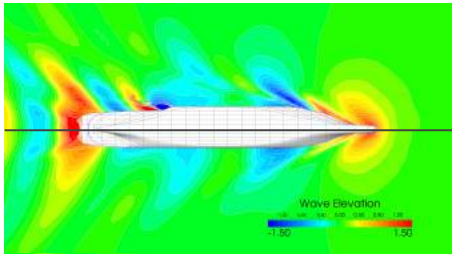
- Total resistance is obtained, by adding a correlation for roughness component and the wind resistance.

- Balancing the total resistance with the propeller thrust, the self-propulsion condition is established yielding the required power for a certain operating condition.

- Fully automated simulation environment of such flow simulations in parallel on high-performance cluster with more than 6000 CPU cores on Linux Platform and using Python scripting

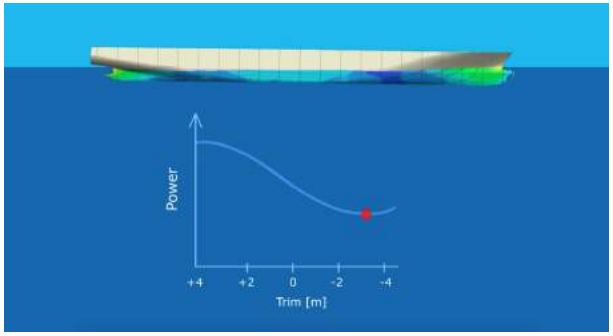
- Hull mesh produced with HexpressTM (around 3000000 cells) and post-processing with CFView

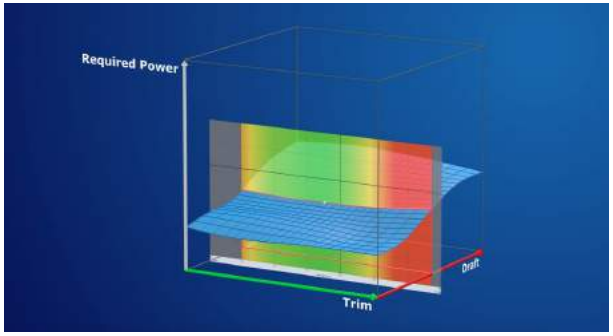




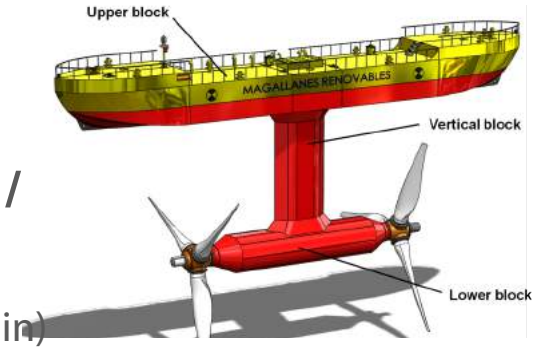
Effective trim optimisation – DNV GL Maritime Advisory

- Many different speed and draft combinations are simulated in a time efficient manner. A typical matrix is 7drafts x7 speeds x7 trims (343 points)
- Hull resistance is calculated in calm water.
- Simulation results are validated against existing towing tank data of tested operating conditions.
- Dynamic sinkage and trim are taken into account.
- Results are presented as a set of hull resistance curves, for a given speed, draft, displacement, on board computer visualization tool for the crew, which is very easy to use.
- Many trim optimisation studies in cooperation with major ship owners, world shipyards and design offices





WORK EXPERIENCE



November 2016 - March 2018

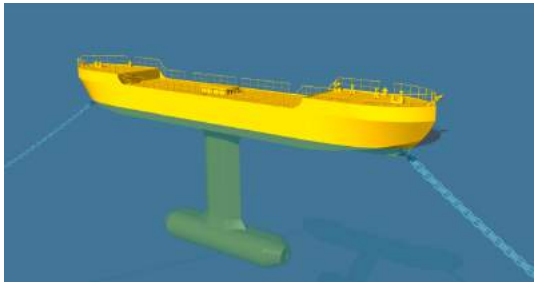
Project Manager (Senior Naval Architect / R&D Engineer)

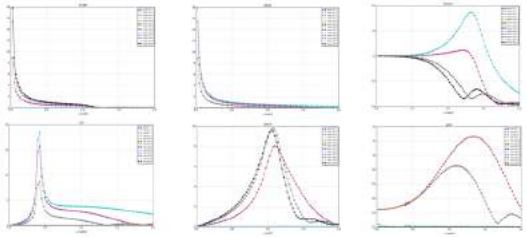
at Sea Master Consulting & Engineering, Cádiz (Spain)

Project Manager dealt with shipbuilding and marine renewable energies projects focused on basic, detailed, production design and engineering solutions.

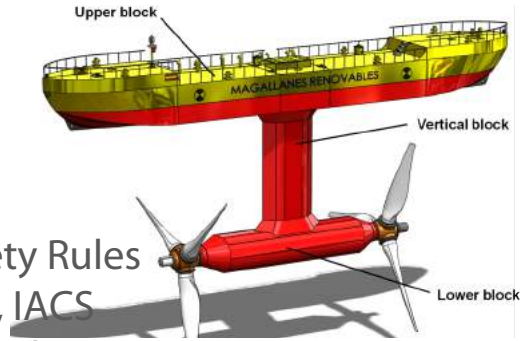
Main Activities:

- Member of the design team a Floating Tidal Generation Conversion Unit
- Project Manager involved in diverse Ferry & ROPAX projects
- PM of the design team for the mechanical piping system of 1500 RoPax Ferry



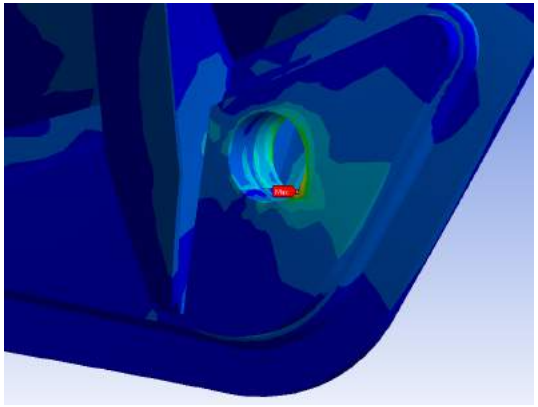


WORK EXPERIENCE



Main Activities:

- Application and verification of Classification Society Rules and Regulations (DNV GL, BV, LR, ABS) and SOLAS, IACS Unified Requirements, EN/ISO/IEC Standards and other recommended practices (API RP 2SK)
- Supervision and approval of technical drawings and calculations as wells as technical support in inclining test
- Development and continuous improvement of production process (Lean Management)

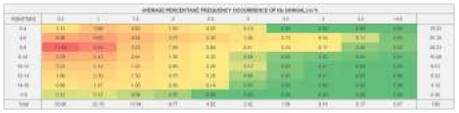


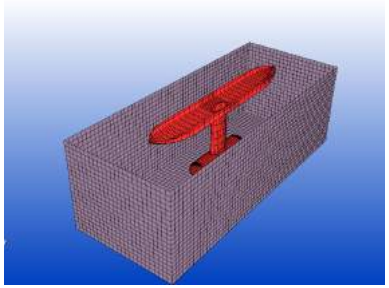
Magallanes Project – Sea Master Consulting & Engineering

Floating Tidal Unit Converser (ATIR Platform):

- Hydrodynamic performance assessment of the device with the study of the seakeeping and the design of a mooring system

- CFD Simulations in full scale using Hydrostar (recognized B.E.M. solver to evaluate 1st & 2nd order wave loads and induced motions) and Ariane (Multi-body mooring software)





Research projects:

- Meshing studies

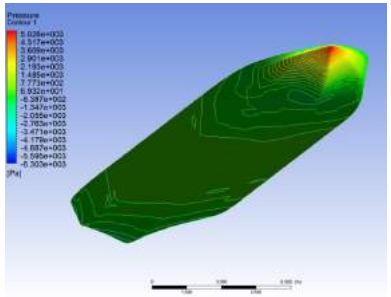
- Validation of drag coefficients (Cx,Cy, Cz) with Sea trial and with a CFD method

- Limit state simulations

- Mooring system design assessment:

  - Rock anchors

  - Adding chain clump weights



Scope of work included:

- Analysis and evaluation of Meteocean data (wave, wind, …)

- Waves predictions based on scatter diagram data to joint probability

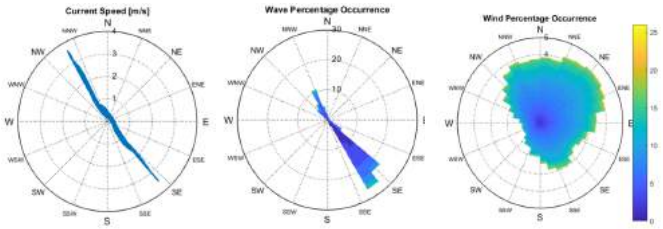
- Evaluation of the loads on the mooring lines with a quasi-dynamic approach to determinate the low frequency response of the moored object and followed by superimpo-sition on the wave frequency motions.

- A 2/4-legged catenary system was performed.

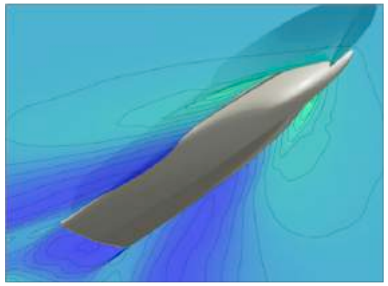
- Definition of the operational profile

- Evaluation of max. excursion under operational condition and survival condition.

- Around 150 Cases extreme condition were studied (max./mean load on anchor and on mooring point at hull)



WORK EXPERIENCE

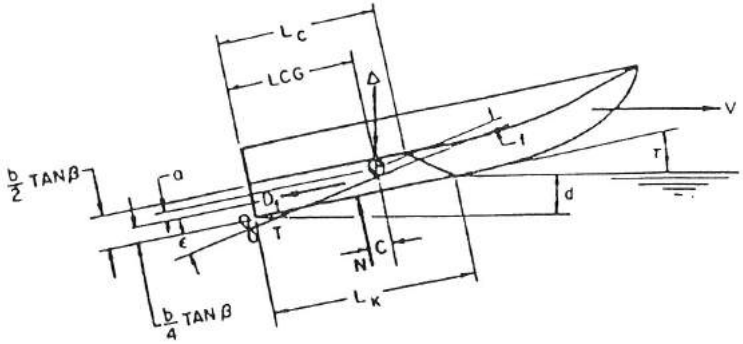


October 2018 - Today

Professional Freelancer (partial-time)

at Navitecture UG, Hamburg

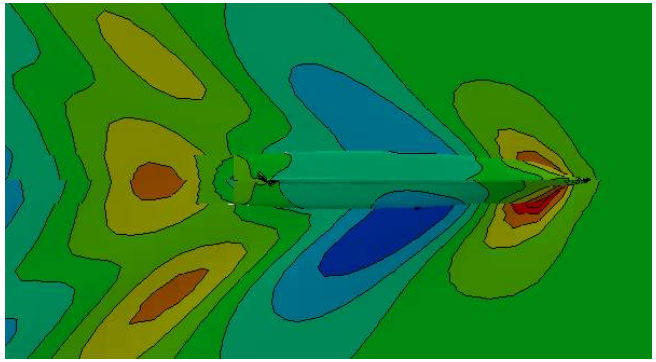
Currently, I am taking part in the design and engineering works of two high-speed crafts (HSC) and one fast-displacement vessel. Today, my main responsibilities are the prediction of resistance and propulsion, as well as the hydrodynamic optimisation and seakeeping performance assessment.



WORK EXPERIENCE

Main Activity:

- Creating general arrangements and preparing all naval architecture calculations (weights, lines plan, stability, …)
- Prediction of resistance and propulsion as well as the seakeeping performance
- Responsible for ship design and layout incl. 2D drawings and 3D models
- Develop and present engineering concepts and market analyses
- Interpretation of international and national applicable classification societies rules and regulations



To be continued...